



Potential and Challenges for Application of Microbiomes in Agriculture

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Introduction

The plant microbiome, which includes a wide range of microorganisms living in association with plants, plays a crucial role in plant health, productivity, nutrition, disease resistance, stress tolerance and even the sensory qualities of crops such as taste and aroma. Understanding how plants interact with their microbiomes requires combining insights from basic research, ecological theory and technological innovation. This article examines the positive impacts of plant-associated bacteria and fungi and highlights the main challenges in using microbiome knowledge for sustainable agricultural practices.

Key Benefits of Plant Microbiomes

Plant Growth Enhancement

Certain bacteria, such as *Bacillus velezensis* GB03, help plants grow by influencing hormone pathways like auxin, cytokinins, ethylene, and gibberellin. These effects are often the result of complex microbial networks rather than individual species, emphasizing the importance of community interactions. Some microbes can also inhibit root growth by producing auxin, showing that balanced hormone regulation is essential for optimal plant structure.

Improved Nutrition

Microbiomes enhance nutrient availability, especially through nitrogen fixation by rhizobia in legumes and phosphorus uptake facilitated by arbuscular mycorrhizal fungi in various plants. While nitrogen fixation in nonlegumes is less efficient than in legumes, it remains a significant area of research. Other mechanisms include microbial-induced changes in root structure and nutrient transporter expression, which improve nitrogen and phosphorus absorption. Iron mobilization by root-associated bacteria is another important nutritional function.

Disease Protection

Microbiomes contribute to plant defense against soil-borne and foliar pathogens through mechanisms such as antibiotic production, competition for resources (like iron), and priming the plant's immune system. Soils rich in beneficial microbes, known as suppressive soils, have been found worldwide. Disrupting pathogen communication and altering microbial community structure are additional strategies for disease prevention. Viral components of the microbiome also affect host-microbe dynamics, with phage therapy showing promise for targeted pathogen control.

Stress Tolerance

Microbes, particularly Actinobacteria like *Streptomyces* and *Variovorax*, help plants withstand drought and salinity stress. They do this by modulating plant hormone signaling, improving water use efficiency, and changing root architecture. However, the effectiveness of these microbes can vary depending on plant genetics and environmental conditions, highlighting the complexity of these interactions.

Flavor and Quality Improvement

Microbial communities influence the flavor profiles of fermented agricultural products such as wine, cheese, and coffee. Microbial interactions shape the metabolite profiles, affecting aroma and taste. For example, microbiomes regulate nitrogen metabolism in tea plants, impacting theanine content, which is crucial for tea quality. These findings suggest opportunities for microbiome-based enhancement of crop and product quality.

Challenges in Agricultural Application

Identification and Delivery of Beneficial Microbes

Although progress has been made in isolating beneficial microbes in controlled environments, applying them in complex agricultural soils presents significant challenges. Native soil microbiomes often resist invasion by introduced strains due to competition and priority effects. Microbes must be carefully selected for compatibility with native communities and host plants to ensure successful establishment. Delivery methods such as seed coating, soil inoculation, and foliar sprays vary in efficiency and scalability, and the choice of carriers affects microbe survival and colonization.

Microbial Invasion and Persistence

Many beneficial strains struggle to persist in natural environments. Host plant genetics, soil factors, and climate strongly influence microbiome assembly. Microbial consortia tend to favor native host preferences, and early colonization enhances community stability against newcomers. Some beneficial microbes can become opportunistic pathogens under certain conditions, underscoring the need to balance microbial community functions for plant health.

Trade-offs and Community Dynamics

Introducing beneficial microbes can affect existing plant-microbe and microbe-microbe interactions, sometimes altering plant growth-defense trade-offs. Microbes that suppress immunity may increase vulnerability to opportunistic pathogens, while immune activation can restrict beneficial microbes. Environmental stresses, nutrient levels, and microbial interactions collectively influence these complex dynamics, requiring integrated knowledge for successful application.

Genetic and Environmental Complexity

Plant genetic variation subtly shapes microbiome composition through root exudates and immune responses. Identifying host genes that modulate microbiome assembly is challenging due to polygenic traits and gene-environment interactions. Molecular breeding or genetic engineering to favor beneficial microbes holds promise but must navigate trade-offs in plant growth and defense.

Use of Multi-omics, Machine Learning, and Experimental Approaches

Advances in genomics, transcriptomics, and metabolomics, combined with machine learning, enable prediction of plant-microbiome interactions and identification of beneficial microbial strains and metabolites. Machine learning models aid in hypothesis generation for microbiome functions but require high-quality data and interpretability. Experimental reductionist approaches complement data-driven insights to unravel mechanisms.

Agricultural Implications and Future Directions

Microbiome-based strategies can reduce reliance on chemical fertilizers and pesticides, promoting sustainable agriculture. Large-scale production and formulation of microbial inoculants face challenges

related to costs, stability and regulation. Engineering crop genotypes for improved microbiome interactions is a long-term strategy. Microbial engineering and synthetic communities offer alternatives but require further research for field efficacy. Using microbial metabolites and informed application within precision agriculture frameworks can enhance microbiome benefits. Overall, interdisciplinary research integrating ecology, molecular biology, agronomy, and data sciences is essential for realizing the full potential of microbiomes in agriculture.

